

DMITRY BERESNEV

ML Engineer & Data Scientist

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📍 Innopolis, Russia

SUMMARY

MSc student in Computer Science specializing in AI and Data Science with research focus on ML optimization, LLMs, and reasoning-oriented learning. Experienced in developing ML and LLM pipelines with PyTorch, fine-tuning and evaluating language models for reasoning tasks, and conducting empirical research. Proven track record leading projects from conception to deployment, including an AI-featured EdTech platform serving 200+ active users. Seeking challenging R&D positions to contribute to state-of-the-art ML and AI solutions.

EDUCATION

MSc in Computer Science

AI & Data Science

📅 2024 – 2026

📍 Innopolis University

- **Thesis:** New and Efficient Facet-Based Identification methods for Rank-Deficient Simplex-Structured Matrix Factorization [in progress]
- **Supervisors:** Valentin Leplat, Nicolas Gillis
- **Relevant Coursework:** High Dimensional Data Analysis, Linear Optimization, Advanced Statistics, Advanced Machine Learning, Reinforcement Learning

BSc in Computer Science

AI & Data Science

📅 2020 – 2024

📍 Innopolis University

- **Thesis:** Text plagiarism detection in the field of large language models using reinforcement learning
- **Supervisor:** Armen Beklaryan
- **Relevant Coursework:** Optimization Methods in Machine Learning, Reinforcement Learning, Natural Language Processing, Practical Deep Learning

RESEARCH EXPERIENCE

Huawei: Wireless Data Transmission

Researcher, ML Engineer

📅 2024 – Present

📍 ISP RAS & Innopolis University

- Designing and simulating Deep AI models for wireless distribution of devices to base stations under time and resource constraints for Huawei
- Developing and implementing models on PyTorch, creating training-testing pipeline, and conducting experiments
- **Stack:** PyTorch, Numpy
- **Supervisor:** Aleksandr Beznosikov

Pyligent: Diligent Learning Python framework

Researcher, ML Engineer

📅 2025 — Present

📍 MSU AI Center

- Implementing and testing Diligent Learning, a novel approach to training and fine-tuning LLMs for reasoning tasks, based on [From Reasoning to Super-Intelligence: A Search-Theoretic Perspective](#)
- Engineering the pyligent framework and research pipeline for reasoning-oriented LLM training and experimentation
- Contributing research ideas for reasoning-oriented LLM training and helping shape new directions within the project
- Conducting experiments with LLM fine-tuning workflows and releasing the Python implementation as a [PyPI package](#)
- **Stack:** vLLM, TRL, Transformers
- **Supervisor:** [Petr Anokhin](#)

Applied AlphaEvolve: CAD Reconstruction

Researcher

📅 Summer 2025

📍 Skoltech Summer School (SMILES-2025)

- Applied OpenEvolve (open-source AlphaEvolve) to CAD reconstruction and combinatorial geometry using LLM-driven evolutionary search
- Achieved optimal ball partition results matching theoretical bounds in dimensions 2–13
- Outperformed zero-shot LLM baselines across multiple complex 3D shapes
- Established comprehensive benchmark pipeline with 7 evaluation metrics
- **Stack:** PyTorch, OpenEvolve, CAD libraries
- **Supervisor:** [Petr Anokhin](#)

Text Plagiarism Detection Using DRL

Researcher

📅 2024

📍 Innopolis University

- Designed novel DRL-based approach for plagiarism detection achieving best MSE of 0.108 on synthetic dataset
- Proposed three architectures based on DQN, A2C, and REINFORCE with best results from REINFORCE model
- **Stack:** PyTorch, Numpy, Pandas
- **Supervisor:** [Armen Beklaryan](#)

INDUSTRIAL EXPERIENCE

Innopolis CIPR

ML Developer

📅 2025

- Designed and implemented RAG pipeline over proprietary Angular frontend repositories
- Built indexers: Inverse Index, BallTree with model-generated embeddings, and partially Faiss
- Connected local generative models and designed full pipeline of scraping, embeddings generation, indexing, and retrieving
- Approved quality on gold queries provided by experts
- **Stack:** PyTorch, Docker, FastAPI, Faiss

Gazprom CPS

ML Engineer

📅 2024

- Designed and trained predictive ML model to identify causes of defects in construction facilities, achieving 80% accuracy on proprietary dataset
- Responsible for full working pipeline: data preprocessing, feature engineering, model building and validation
- **Concepts:** Tree-based models, MLP-based models, ensemble methods, Transformers
- **Stack:** PyTorch, Numpy, Scikit-learn, Pandas

Advanced Engineering School IU

ML Developer

📅 2023

- Developed code generation model using transformer-based architecture
- Fine-tuned **Gorilla** model on proprietary dataset
- Contributed significantly to research
- **Concepts:** LLMs, Transformers, LoRA
- **Stack:** PyTorch, Numpy, Pandas

SELECTED PROJECTS

Accept School

Founder, CEO

📅 2023 – Present

- Led full-stack design of comprehensive EdTech platform combining ML with modern web technologies, currently utilized in educational organizations with approximately 200 active users
- Developed code plagiarism detection system using ML and implemented generative AI for hint suggestions, text and image generation using open-source LLMs
- Defined development and operational processes, engineered backend with FastAPI and MongoDB, built frontend with Next.js
- **Stack:** PyTorch, FastAPI, Next.js, MongoDB, Docker, Apache Kafka

DoWell

ML Developer, Tech Leader

📅 2025

- Designed and implemented RAG architecture for intelligent conversational system simulating expert consultations across professional domains
- Deployed and connected generative models, engineered backend using FastAPI
- **Stack:** PyTorch, FastAPI, Docker

Detecting AI-generated Python Code via ML

Developer

📅 2025

- Achieved 95.9% accuracy with CodeBERT model on synthetic dataset
- Developed efficient AST-based Random Forest achieving 83.5% accuracy with 2ms inference time
- Engineered dataset generation pipeline using 4 LLMs with specialized prompts
- Integrated LIME explainability framework for model interpretation
- **Stack:** PyTorch, Transformers, Tree-sitter, LIME

PyFinder: Python Documentation Search

Developer

📅 2025

- Built Information Retrieval system combining traditional inverted indexing with LLM-powered semantic search
- Performance: $F1@1=0.53$, $nDCG@1=0.83$ with LLM embeddings + Ball Tree indexer
- Implemented RAG pipeline with prompt engineering and context retrieval
- **Stack:** PyTorch, sentence-transformers, FastAPI, Next.js

PROFESSIONAL DEVELOPMENT

Skoltech Summer School (SMILES-2025)

Harbin, China

📅 2025

12-day intensive course focused on generative methods in AI, organized by Skoltech in collaboration with the Harbin Institute of Technology (HIT), China.

- **Result:** Defense of project 'Applied AlphaEvolve: CAD Reconstruction and Combinatorial Geometry'
- **Concepts:** Generative AI, LLMs

Course 'LLMs-based Agents'

Skoltech

📅 2025

Professional development program organized by Skoltech and devoted to modern technologies in the field of Agentic AI. Covered topics: ReAct, RAG, LlamaIndex, LangChain, n8n.

- **Result:** Lawyer Agent featured with RAG and Web Tools based on LlamaIndex framework
- **Concepts:** Agentic AI, LLMs

Course 'Technologies for ensuring information security and data transmission in UAS'

UAVPROF

📅 2025

Professional development program organized by UAVPROF and devoted to data transmission and information security in Unmanned Aircraft Systems (UAS)

- **Result:** Implemented secure base-drone data transmission via ViPNet protocol

TEACHING EXPERIENCE

Teaching Assistant

MSU Faculty of AI Winter School

📅 2026

Organized and delivered tutorials for students on LLM reasoning topics, supporting educational activities of the school.

Teaching Assistant

Innopolis University

📅 2025

Teaching assistant for Introduction to Optimization course for 2nd year bachelor students. Conducting tests and laboratory work.

Teaching Assistant

Yandex Student Camp on Math in AI

📅 July 2024

📍 Innopolis

Designed and provided materials for seminars and homeworks on 'Optimization Methods in Machine Learning' course under guidance of **Aleksandr Beznosikov**

SKILLS

ML & Data Science

PyTorch, TRL, vLLM, Scikit-learn, PuLP, JAX

Programming Languages

Python, TypeScript, C/C++, Rust

Tools & Platforms

Docker, Git, LaTeX, Postman

Web & Databases

FastAPI, Next.js, Astro, PostgreSQL, MongoDB

LANGUAGES

Russian



English

